23CSE202 DATABASE MANAGEMENT SYSTEMS
S3 B.Tech CSE
Department of Computer Science and Engineering
Amrita School of Computing, Amritapuri Campus

3. Database Design Document

Group No: B1
Team Members:

Roll No	Student Name
AM.SC.U4CSE25117	G.NEHA ABHIGNA
AM.SC.U4CSE25138	NEHA B NAIR
AM.SC.U4CSE25143	P.SRINIJA
AM.SC.U4CSE25166	JANWI YADAV

Project Title: Grey Lore

3.1. Detailed Description of the Project

Graylore is an OTT video streaming platform that allows users to register, create profiles, subscribe to plans, browse content, watch movies/shows, maintain watch history, and rate content. An administrator can manage content, licenses, subscriptions, users, and royalty payouts.

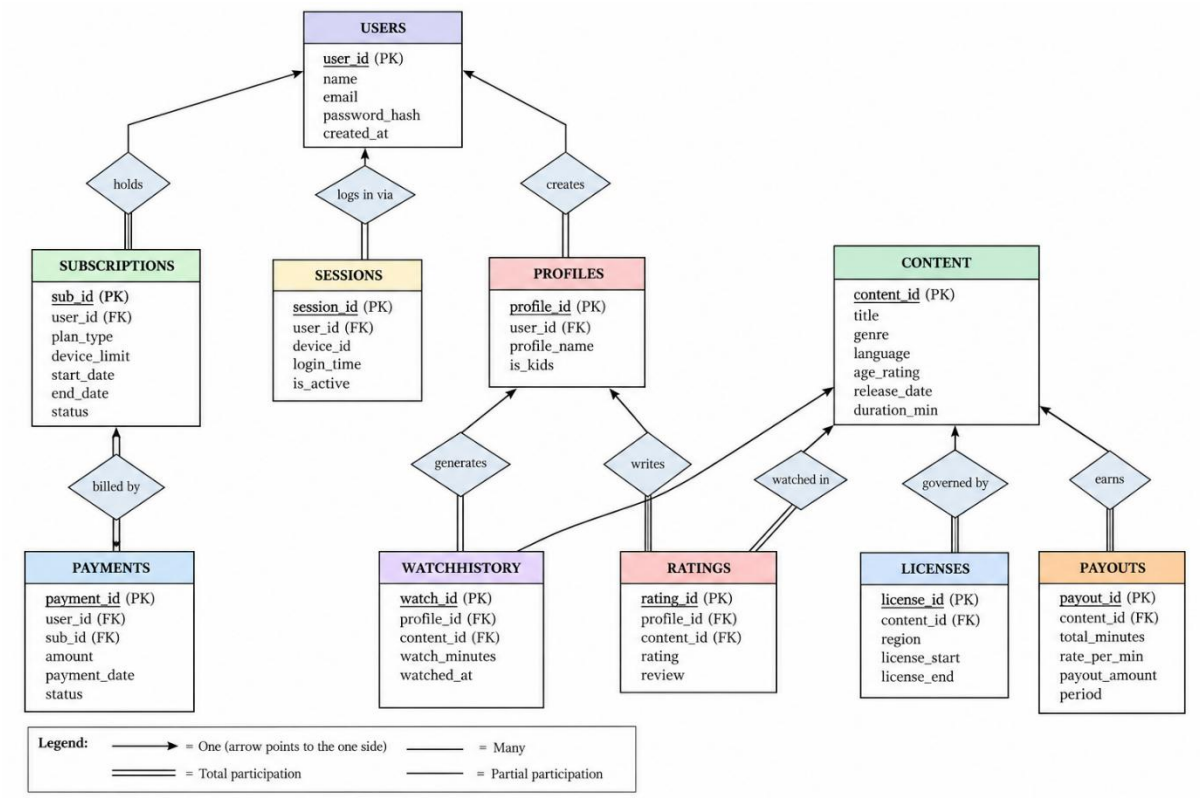
A User can have multiple Profiles, Subscriptions, and Sessions. Profiles maintain separate viewing activity and can be marked as kids profiles. Each subscription stores its plan type, device limit, validity period, and status, while Payments records subscription transactions.

The platform stores Content along with its genre, language, age rating, release date, and duration. Licenses determine the regions and time periods in which each content title is available.

When a profile watches content, the activity is stored in WatchHistory, including watch duration and viewing time. Profiles can also submit Ratings and reviews for content. Payouts record royalties for content based on total watch minutes and the applicable rate.

The database uses primary and foreign keys to maintain relationships and data integrity between these entities.

3.2. ER Diagram



3.3 Relational Schema

Users(user_id PK, name, email, password_hash, created_at)

Subscriptions(sub_id PK, user_id FK → Users, plan_type,
device_limit, start_date, end_date, status)

Profiles(profile_id PK, user_id FK → Users, profile_name, is_kids)

Content(content_id PK, title, genre, language, age_rating,
release_date, duration_min)

Licenses(license_id PK, content_id FK → Content, region,
license_start, license_end)

Sessions(session_id PK, user_id FK → Users, device_id,
login_time, is_active)

WatchHistory(watch_id PK, profile_id FK → Profiles,
content_id FK → Content, watch_minutes, watched_at)

Ratings(rating_id PK, profile_id FK → Profiles,
content_id FK → Content, rating, review)

Payments(payment_id PK, user_id FK → Users,
sub_id FK → Subscriptions, amount, payment_date, status)

Payouts(payout_id PK, content_id FK → Content,
total_minutes, rate_per_min, payout_amount, period)

3.4 Normalization and Functional Dependency Analysis

Functional Dependency Analysis:
Users:
user_id → name, email, password_hash, created_at

Subscriptions:
sub_id → user_id, plan_type, device_limit, start_date, end_date, status

Profiles:
profile_id → user_id, profile_name, is_kids

Content:
content_id → title, genre, language, age_rating, release_date, duration_min

Licenses:
license_id → content_id, region, license_start, license_end

Sessions:
session_id → user_id, device_id, login_time, is_active

WatchHistory:
watch_id → profile_id, content_id, watch_minutes, watched_at

Ratings:
rating_id → profile_id, content_id, rating, review

Payments:
payment_id → user_id, sub_id, amount, payment_date, status

Payouts:
payout_id → content_id, total_minutes, rate_per_min, payout_amount, period

These functional dependencies indicate that each primary key uniquely determines all the non-key attributes of its corresponding relation. They help maintain data consistency and reduce redundancy in the database.

Candidate Keys and Attribute Closure:

The primary key of each relation is a candidate key because its attribute closure contains all attributes of that relation.

For example, for the Content relation,
content_id → title, genre, language, age_rating,
release_date, duration_min

The attribute closure is,
content_id+ = {content_id, title, genre, language,
age_rating, release_date, duration_min}

Since the closure contains all attributes of the Content relation, content_id is a superkey. Since it is minimal, it is also a candidate key and is selected as the primary key.

The candidate keys for the relations are,

Relation	Candidate Key
Users	user_id
Subscriptions	sub_id
Profiles	profile_id
Content	content_id
Licenses	license_id
Sessions	session_id
WatchHistory	watch_id
Ratings	rating_id
Payments	payment_id
Payouts	payout_id

Normalization:

The Graylore database is normalized through the following stages,

First Normal Form (1NF),
All relations contain atomic values and do not have repeating groups. Each record is uniquely identified by a primary key. For example, each viewing activity is stored as a separate record in the WatchHistory relation.

Second Normal Form (2NF),
All relations are in 1NF and there are no partial dependencies because the relations use single-attribute primary keys. Therefore, every non-key attribute is fully dependent on the complete primary key.

Third Normal Form (3NF),
All non-key attributes depend directly on the primary key, and there are no transitive dependencies between non-key attributes. For example, in the Content relation, all attributes depend directly on content_id.

Boyce-Codd Normal Form (BCNF),
The relations satisfy BCNF because, for every non-trivial functional dependency, the determinant is a candidate key or superkey. Therefore, the database design reduces redundancy and helps prevent insertion, deletion and update anomalies.

Hence, the Graylore database is normalized up to BCNF.